

```

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$GPGSA,A,3,19,28,11,13,24,10,08,23,,,1,13,1.02,0.86*07
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$GPGSV,4,1,14,13,64,172,23,07,56,359,,19,44,061,34,28,37,263,18*7F
$GPGSV,4,2,14,08,34,322,15,23,29,158,27,11,22,135,19,03,18,043,*73
$GPGSV,4,3,14,10,17,258,24,24,07,138,26,06,07,039,,17,06,204,*77
$GPGSV,4,4,14,05,04,301,,34,,,*48
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$GPGGA,164804.000,2656.9030,N,07544.6624,E,1.8,1.02,179.4,M,-42.2,M,,*73
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$GPRMC,164804.000,A,2656.9030,N,07544.6624,E,0.00,60.23,040311,,,A*58
$GPVTG,00.23,T,M,0.00,N,0.00,K,A*0A
    
```

Fig. 2: Sample data from GPS receiver in NMEA protocol

TABLE-I
NMEA format of latitude and longitude

A.	xxmm.dd	Latitude
		xx = degrees mm = minutes dddd = decimal part of minutes
B.	yyymm.dddd	Longitude
		yyy = degrees mm = minutes dddd = decimal part of minutes

Latitude and longitude conversion

Conversion to decimal system:
 Latitude = $xx + (mm.dd/60)$
 Longitude = $yyy + (mm.dddd/60)$

TABLE-2
Baud rates for NMEA

Baud Rate	Characters/Sec
300	30
1200	120
2400	240
4800	480
9600	960
19200	1920
57600	5760
115200	11520

Note: 4800 and 9600 are common baud rates for GPS receivers.

Software tools for GPS receivers

Various software tools are available for GPS receivers. These tools communicate with receivers, read the NMEA messages and give the required output.

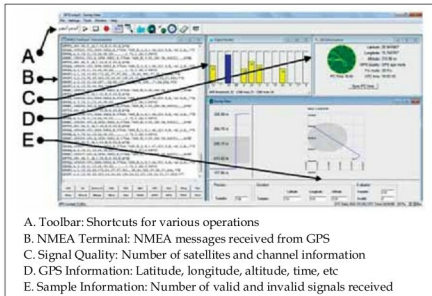


Fig. 3: GPS Cockpit

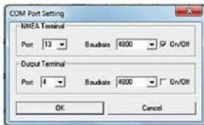


Fig. 4: Comport setting in GPS Cockpit

For example, latitude, longitude, altitude, numbers of satellite, etc.

GPS Cockpit is one of the software that is used widely. Here, we see how to use it.

How to connect

To connect a device follow the steps mentioned below:

1. Connect the device to the appropriate serial/USB port.
2. Select connect button from toolbar.
3. Select 'Port' and 'Baudrate' (4800/9600 generally) in 'NMEA Terminal.' Do not select any output terminal if you do not want to use NMEA messages further.
4. Wait till GPS receiver responds. It may take 10-15 minutes.
5. For disconnecting select button from toolbar.

Navigation and GPS

Navigation is the most important application of GPS. The process of monitor-